TEXORA · LECTURE SHEET

Circular Interlock Machine

Interlock gating, two lengths of needle in each bed, and two cam systems working alternately: the interlock machine knits two ribs into one locked fabric, smooth on both sides — at the cost of feeders, speed, and a timing discipline stricter than the rib machine's.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 6 · LEVEL 3 TERM 1

1. Gating and needle arrangement

Unlike rib gating, **interlock gating** places each dial needle *directly behind* a cylinder needle (needle-to-needle). Each bed carries two needle sets alternating around the bed: **long** and **short** (distinguished by butt position/length). Two cam systems serve the beds: **System A** knits long cylinder needles with short dial needles; **System B** knits short cylinder needles with long dial needles. At System A feeders the B-needles are held out (miss); at System B feeders the A-needles are held out. One complete interlock course therefore needs **two successive feeders**, and consecutive feeders alternate A, B, A, B around the machine — feeder count for interlock must be even, and fabric length output per revolution is F/2 courses.

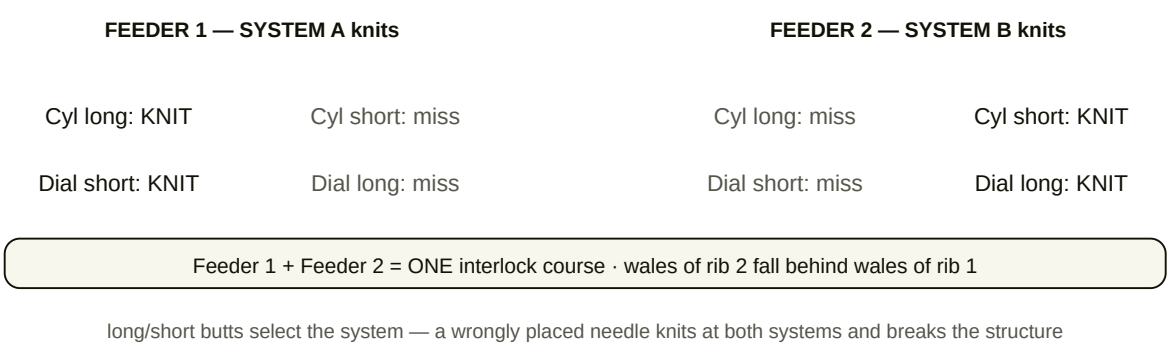


Figure 1 — Interlock needle selection. Two complementary half-courses, knitted at alternate feeders, intermesh into one course with coincident wales.

2. Cam systems and timing

Each bed has two stitch-cam tracks (for long and short butts) whose settings must match pair-wise (A-cylinder with A-dial), or the two half-courses differ in loop length and the fabric shows horizontal streaks every other course. Timing is normally **synchronised or slightly delayed**; because the beds are

gated needle-to-needle, vertical clearance between hook and opposing needle is small, so dial height and cylinder/dial concentricity are critical — a high spot on the dial prints as a repeating fault once per revolution. Many interlock frames are convertible: re-gating the dial by half a pitch and fitting rib cams converts the machine to rib work (and vice versa), which is why factories buy "rib/interlock" frames — but conversion is a workshop job, not a shift job.

3. Production — worked example (verified)

WORKED EXAMPLE — INTERLOCK LENGTH AND MASS

Machine: D = 30 in, E24, F = 48 feeders, 20 rpm, stitch length 4.6 mm, yarn 19.7 tex, efficiency 85 %. Relaxed CPC (interlock courses) = 16/cm.

Needles per bed N = π × 30 × 24 = **2262**. But only half the needles of each bed knit at any feeder: yarn per feeder-course = 2 beds × (2262 ÷ 2) × 4.6 mm = 2262 × 4.6 mm = 10.41 m.

Feeder-courses/min = 48 × 20 = 960. Yarn rate = 960 × 10.41 = 9994 m/min = 599,616 m/h. Mass = 599.6 km × 19.7 g/km = 11.81 kg/h; × 0.85 = **10.0 kg/h**.

Interlock courses/min = 960 ÷ 2 = 480. Length = 480 × 60 ÷ 16 courses/cm = 1800 cm/h = 18.0 m/h; × 0.85 = **15.3 m/h**.

Lesson: interlock's kg/h looks low because the fabric is dense and the course rate is halved — compare machines on m²/h and fabric value, not kg/h alone.

4. Quality control points

- **Needle order:** one long needle placed in a short position knits at the wrong system — mixed wales, holes; check the full circle after any needle change.
- **Pair-matching of stitch cams:** measure loop length from both systems separately (unrove one course of each) and equalise.
- **Dial concentricity/height:** once-per-revolution repeats point here, not to feeders.
- **Yarn:** interlock's double structure hides yarn variation less than jersey at the surface — even, low-hairiness yarn pays; elastane interlock needs plating discipline at both systems.

5. Interlock derivatives on the same frame

By selection: **punto di Roma** (interlock variant with single-bed courses in the repeat — lighter, more stable), **eight-lock**, **interlock piqué and twill effects**, and striped interlock by colour feeders in A/B pairs (a colour change must occupy a complete A+B pair or the stripe splits across half-courses).

Glossary

Interlock gating — dial needles directly behind cylinder needles. **Long/short needles** — alternating needle sets selected by butt position for Systems A and B. **Half-course** — the rib course knitted at one feeder; two make one interlock course. **Punto di Roma** — interlock-family double fabric with plain single-bed courses in the repeat.

Key points

• Interlock = needle-to-needle gating + long/short selection + two alternating cam systems; one course = two feeders.
• Verified output (30 in/E24/48F/20 rpm): 10.0 kg/h, 15.3 m/h — judge on area and value.
• Pair-matched stitch cams and correct needle order are the quality core.

Practice

1. A 36-feeder interlock machine runs at 22 rpm. How many interlock courses per minute does it knit, and how many metres per hour at CPC 15/cm and 88 % efficiency?
2. Why must a colour stripe change occupy an A+B feeder pair?
3. A fault repeats exactly once per machine revolution on an interlock frame but moves feeder to feeder on inspection. Where do you look first, and why not at the feeders?
4. State two reasons interlock uses finer gauges (E24–E32) more often than rib.

Answer key

1. Courses/min = 36 × 22 ÷ 2 = **396**. Length = 396 × 60 ÷ 15 = 1584 cm/h = 15.84 m/h; × 0.88 = **13.9 m/h**.
2. One visible course is built from two half-courses. Changing colour at only one feeder of the pair puts the new colour in half the loops of the course — a split, mixed stripe line instead of a clean boundary.
3. At the dial/cylinder geometry — dial height, concentricity, a damaged needle or trick at one angular position. A feeder fault would repeat every feeder-pass (many times per revolution), not once per revolution.
4. The structure is dense and smooth-faced, aimed at fine apparel (polos, underwear), so fine yarns and gauges suit it; also each needle knits only every other feeder, so the per-needle duty is lower and finer needles survive. (Either reason developed earns credit.)